Agenda

* Linear Regression Implementation
* How feature scaling helps easier model training
* Adjusted R-squared
* Introduction to statsmodel
* Assumption of Linear Regression

=✓✓ Assignments

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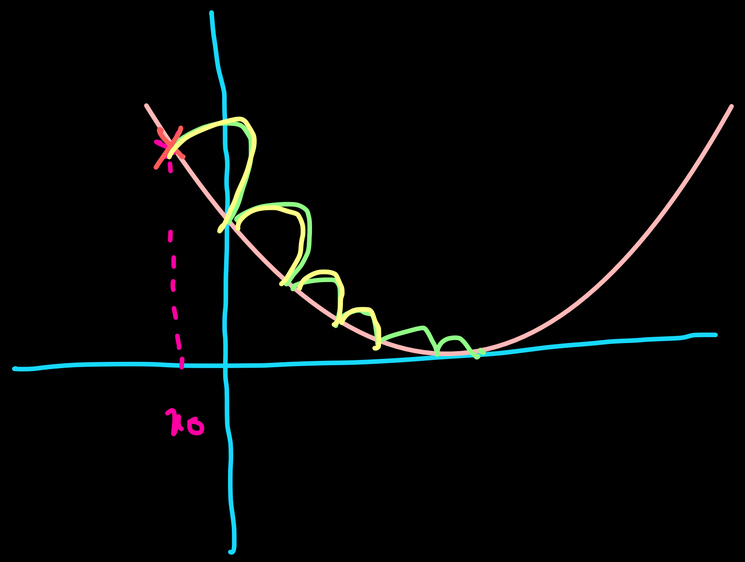
Model Interpretability

⇒ feature importance: → (weights)

⇒ feature scaling :

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1 Gradient Descent Agorithm



1. Pick randomly x_0
2. $\frac{\partial L}{\partial x}$
3. update $x = x - \eta \frac{\partial L}{\partial x}$ (learning rate)

Loss Function

MSE : $L = \frac{1}{n} \sum_{i=1}^{n} (y^{(i)} - \hat{y}^{(i)})^2$

Apply G.D

1. Randomly initialize all 'w vector $[w_1, w_2 \dots w_d]$

** 2. Calculate the gradient $\frac{\partial L}{\partial w_j}$

3. Repeat { $w_j = w_j - \eta \left\{ \frac{\partial L}{\partial w_j} \right\}$ } learning

50
100
200

Calculate the Gradient

$$\Rightarrow \hat{y} = w_1 x_1 + w_2 x_2 + \dots + w_d x_d$$

Loss fn $\leftarrow L = \frac{1}{n} \sum_{i=1}^{n} (y^{(i)} - \hat{y}^{(i)})^2$ → Keep away

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$$\frac{\partial L}{\partial w_j} = \frac{\partial}{\partial w_j}(y - \hat{y})^2$$

This is a function of $\hat{y}$ and $\hat{y}$ is function of w_j

→ apply the chain Rule

$$= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_j}$$

$$= \frac{\partial}{\partial \hat{y}}(y-\hat{y})^2$$

$$= 2(y-\hat{y})(-1) * \frac{\partial}{\partial w_j}(w_1x_1 + w_2x_2 + \cdots + w_jx_j + \cdots w_dx_d)$$

$$\frac{\partial L}{\partial w_j} = 2(\hat{y}-y) * x_j$$

→ for a single data point

For all datapoints

$$\frac{\partial L}{\partial w_j} = \frac{1}{n}\sum_{i=1}^{n} 2\left(\hat{y}^{(i)} - y^{(i)}\right) \times x_j^{(i)}$$

$$\Rightarrow \text{gradient } \frac{\partial L}{\partial w} = \frac{2}{n}\sum_{i=1}^{n} \underbrace{\left(\hat{y}^{(i)} - y^{(i)}\right)}_{A} \cdot \underbrace{x_j^{(i)}}_{B}$$

$A \rightarrow (y_p - y)$

$B \rightarrow x$

$$= \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \\ \vdots \\ \frac{\partial L}{\partial w_d} \end{bmatrix}_{d \times 1}$$

$$\Rightarrow \frac{2}{n}\left(x^T \cdot (y_p - y)\right)$$

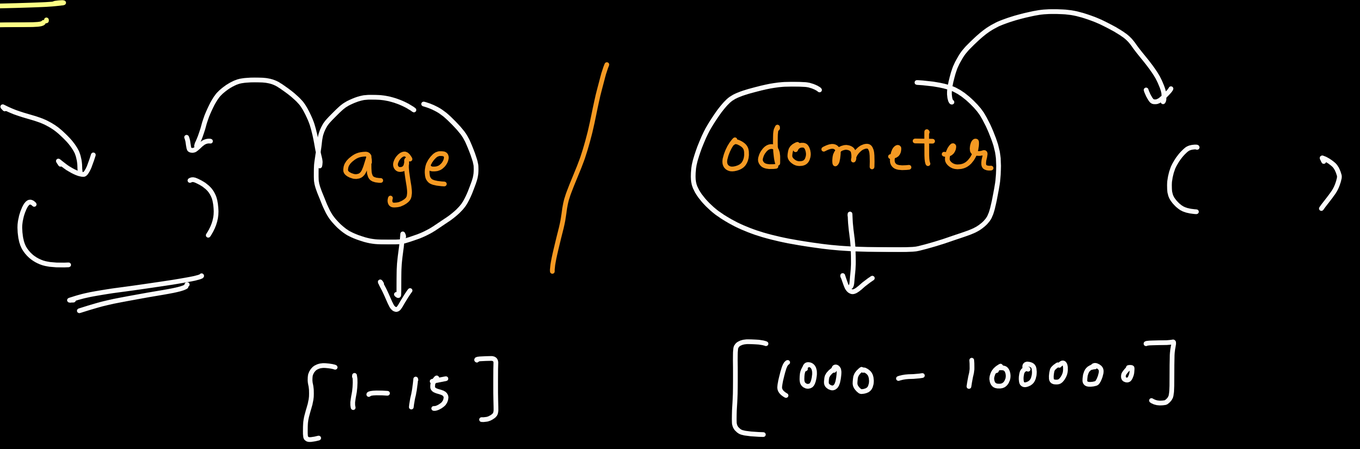
(d, n) $(n, 1)$

$\Rightarrow (d, 1)$ ✓✓

$$R2\text{-score} = 1 - \frac{SS_{res}}{SS_{total}} \rightarrow \frac{\frac{1}{n}\sum_{i=1}^{n}\left[y^{(i)} - \hat{y}^{(i)}\right]^2}{\frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - \bar{y}\right)^2}$$

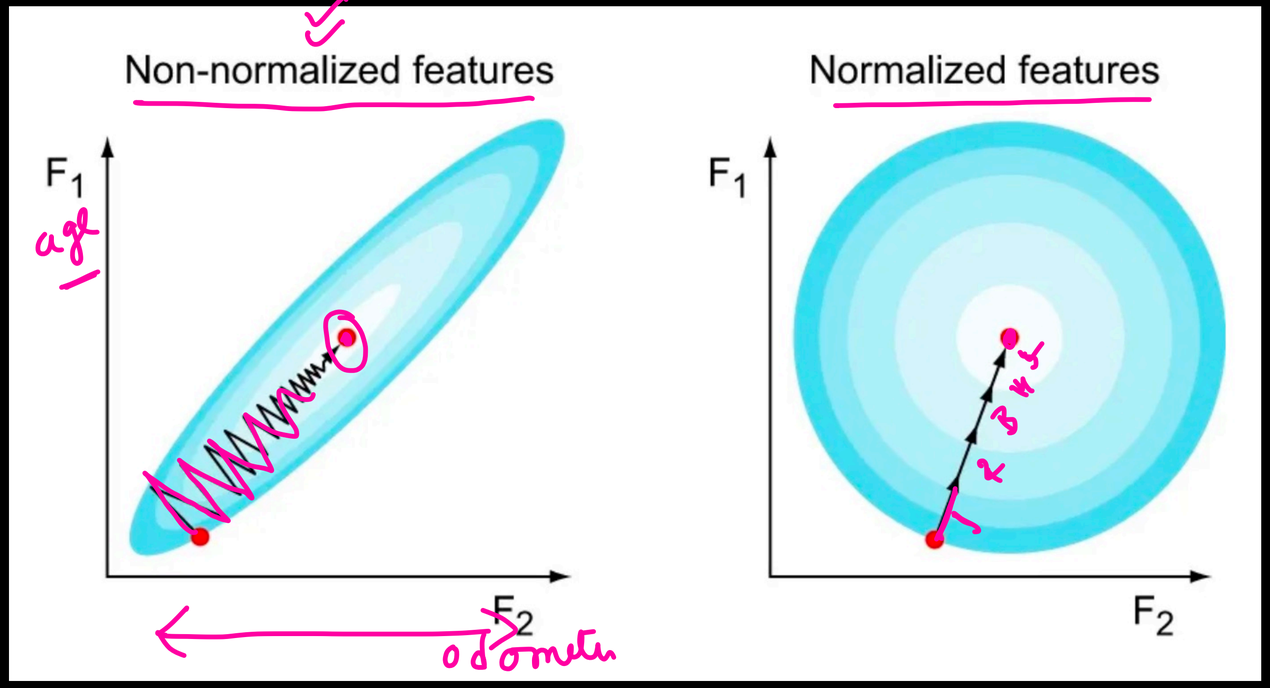
Feature Scaling

(1) Importance



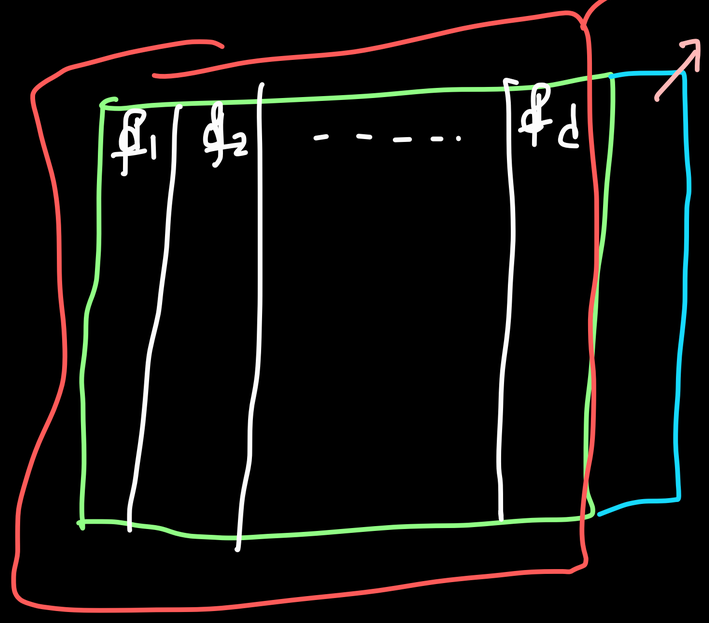
feature scaling must

(2) ✓ G.D converges fast


Non-normalized features
Normalized features
F1
F2
F1
F2

Adjusted R2-Score

$$R2\text{-score} = 1 - \frac{SS_{res}}{SS_{total}}$$



R2-score

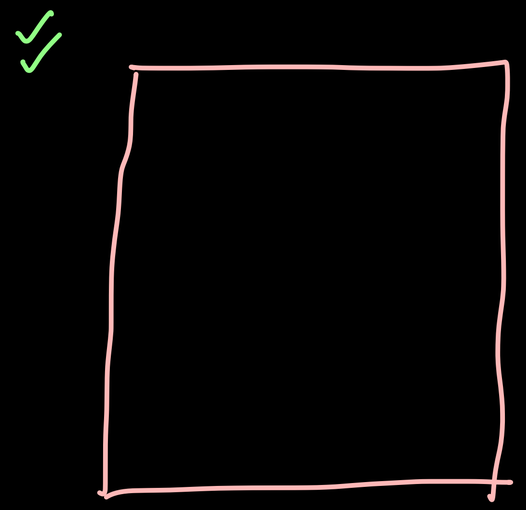
d+1

R2-score

↑ S

↓ S

Remain same



① ✓✓ feature relevant → ✓✓ R2-score ↑s

88% → 92%.

② ✓✓ feature not relevant → R2-score

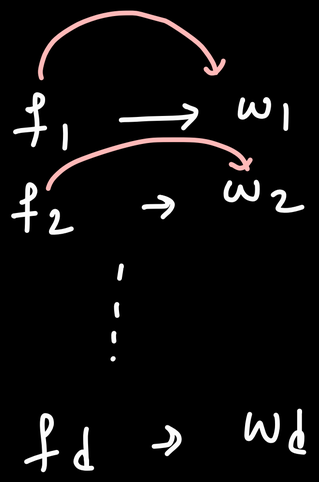
0.88 → 0.89

- ✓ remain same (↓)
- ✓✓ R2-score ↑

0.88 → 0.9

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✓ Why R2-score value is increasing eventhough added an irrelevant feature ?



$w_j \rightarrow 0$

$w_1 x_1 + w_2 x_2$

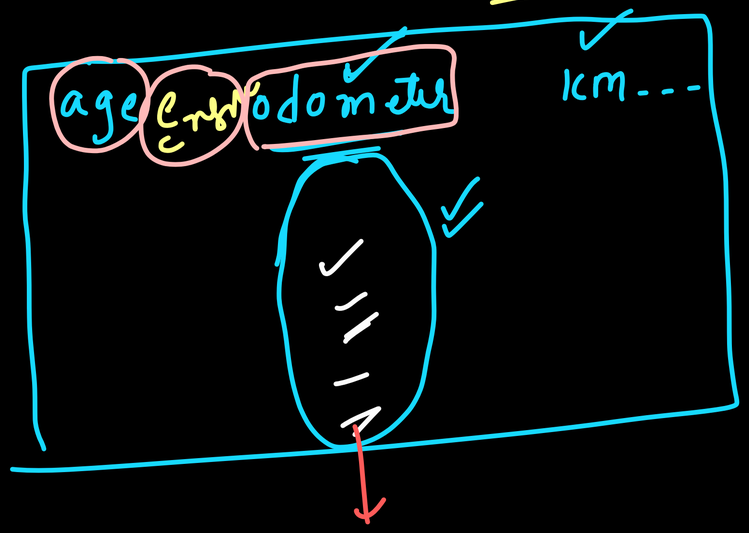
1 extra feature

$w_1 x_1 + w_2 x_2 + w_3 x_3$, $w_3 \rightarrow 0$

R2-score 0.88

✓✓ R2-score value remain same ✓

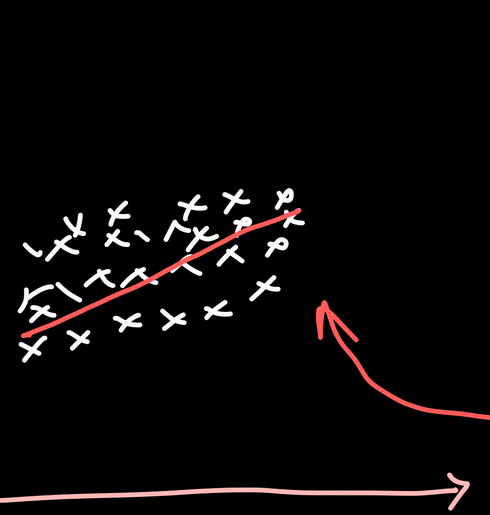
R2 - score ↑s



odometer ↑ → price ↓s

engine ↑ → price ↑

price



=> some slight +ve correlati

0.885 → 0.886

$f_1 \rightarrow w_1$

$f_2 \rightarrow w_2$

$f_f \rightarrow w$

random feature → 0.84

$w_j = 0.4$

$\Rightarrow$ R2-score never decreases, it always increase (or) Remain same

$$R2 = 1 - \frac{SS}{SS}$$

Adjusted R2 score $= 1 - \left[\frac{(1 - R^2)(n-1)}{(n-(d+1))} \right]$

R2-score

$d \rightarrow d+1$

- $n \rightarrow$ no of data
- $d \rightarrow$ dimentions features

case 1: if $d+1$ irrelevant feature

Adj - R2 $\downarrow$

$$d \propto \frac{1}{\text{Adj R2}}$$

n: $10 - 4 \rightarrow 6$

$10 - 5 \rightarrow 5$

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Case 2:

d+1 → relevant feature

R_2

↓

n-

How does the adjusted R2 score differ from the regular R2 score?

16 users have participated

A	The adjusted R2 score accounts for the number of predictors in the model.	38%
B	The adjusted R2 score is always higher than the regular R2 score. ✗	31%
C	The adjusted R2 score considers only the explanatory power of the model.	19%
D	The adjusted R2 score is not influenced by the sample size.	12%

R2-score ↓ ✓ [0-1]

Adj R2-score

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Theoritical

$R2 \rightarrow (-\infty, 1]$

adj $R2 \rightarrow (-\infty, 1]$

Practical / Real world

$[0, 1]$

adj $R2 \rightarrow [0,1] \cup$

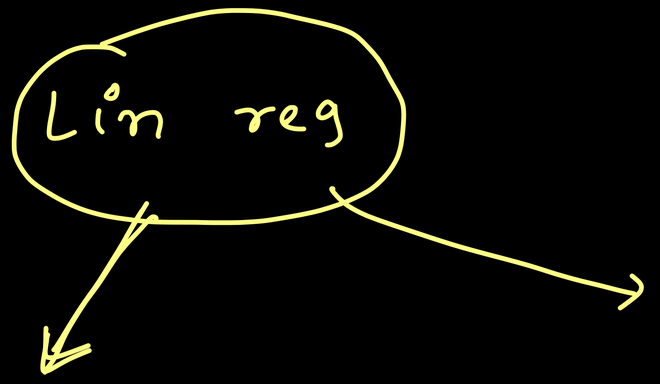
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Stats model

Statisticians

1. Statistians

2. ML



prob & stat

Algebra & optimization

(geometric)

OLS → ordinary least squares

↳ MSE

Why might the adjusted R2 score be considered more reliable than R2 when adding more predictors to a model?

3 users have participated

A	Because it always increases with more predictors.	67%
✓ B	Because it penalizes the model for adding predictors that don't improve the model.	33%
C	Because it is easier to calculate.	0%
D	Because it always equals the R2 score..	0%

End Quiz Now

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